SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology

Course: ABE 721 (AGE 721) - Agricultural Systems Modelling and Simulation

TEST 1

Instructor: Kadeghe Fue, PhD, P.Eng(T)
Time Allowed: 3 Hours

Instructions to Candidates:

- This is an **OPEN-BOOK** examination.
- **NO ELECTRONIC DEVICES** (calculators, laptops, smartphones) are permitted.
- Do not recite definitions. Questions require the synthesis of theory to solve novel engineering scenarios.
- For questions requiring algorithmic implementation, you are expected to write the structural logic of the Python scripts by hand. Syntax perfection is secondary to logical accuracy.
- Answer **ALL** questions in the provided booklet.
- Show all logical steps in your mathematical derivations and explicitly define all variables used.
- Total marks: 100.

Section A: Conceptual Paradigms and Mathematical Formulation (30 Marks)

Question 1: Architectural Trade-offs in Novel Scenarios (10 Marks)
A smart-irrigation startup wants to predict transient volumetric soil moisture (θ) at a 50cm root depth. They possess 5 years of surface weather data and soil moisture sensor logs from a specific sandy-loam farm.

a) Formulate a hypothetical architectural structure for a purely data-driven approach (e.g., a Neural Network) versus a physics-based approach (e.g., utilizing the Richards equation). (4 Marks)

b) The startup suddenly expands operations to a neighboring region characterized by heavy clay soils and unprecedented drought conditions not seen in their 5-year dataset. Critically evaluate why one of your proposed architectures is highly likely to fail catastrophically in this new environment, and how the other architecture can adapt. (6 Marks)

Question 2: State-Space Formulation of a Grain Silo (20 Marks)
You are modelling a continuous-flow grain drying silo. Wet grain flows in continuously at a rate $\dot{m}_{in}$, hot air is blown through it to evaporate moisture, and dry grain flows out at $\dot{m}_{out}$. You are tasked with modelling the total transient mass of liquid water $W(t)$ stored within the silo.

a) Define your strict system boundary. Identify the primary State Variable and at least two Exogenous Inputs dictating the drying dynamics. (4 Marks)

b) Derive the fundamental mass-balance Ordinary Differential Equation (ODE) for $\frac{dW}{dt}$. Introduce and explicitly define a specific parameter representing the thermodynamic drying efficiency of the hot air. (6 Marks)

c) Contrast your derived continuous ODE formulation with a Discrete-Event Simulation (DES). If you were instead modelling the queue of delivery trucks waiting outside the silo, why would the ODE approach become mathematically inappropriate? (5 Marks)

d) **Python Implementation:** Write a Python function `def silo_moisture_derivative(W_current, t, params):` that structurally calculates and returns your derived derivative $\frac{dW}{dt}$. Assume `params` is a dictionary containing all necessary flow rates and efficiencies. (5 Marks)

Section B: Numerical Execution and Parameter Estimation (35 Marks)

Question 3: Numerical Instability in Asymmetric Growth (15 Marks)
You are modelling a rapid fungal pathogen outbreak on a crop canopy using the asymmetric Gompertz growth curve: $\frac{dN}{dt} = rN \ln(\frac{K}{N})$, where N is infected leaf area.

a) Critically analyze why utilizing a large time step (h) with the explicit Forward Euler method is particularly dangerous and prone to overshoot specifically around the inflection point of this asymmetric curve, compared to a linear decay model. (4 Marks)

b) Suppose strict biological constraints mandate that the simulated infected area N can never mathematically drop below zero or exceed K. How does utilizing an implicit approach (like Crank-Nicolson) mitigate the explosive algorithmic instability inherent to explicit methods when CFL limits are violated? (5 Marks)

c) **Python Implementation:** Write a Python function `def heuns_fungal_step(N_current, h, r, K):` that executes and returns a single time step advancement of the fungal population using Heun's Method (predictor-corrector logic) applied specifically to this Gompertz ODE. (6 Marks)

Question 4: Metaheuristics and Custom Objective Functions (20 Marks)
You are calibrating a complex, non-linear tractor draft force equation. The standard Levenberg-Marquardt Algorithm (LMA) continually fails to converge on the global optimum because the error surface is highly rugged due to sudden localized soil compactions.

a) In machinery operations, under-predicting draft force is disastrous (causing engine stalls), while over-predicting is merely inefficient. Deviate from the standard Sum of Squared Errors (SSE) and mathematically derive a custom Objective Function $J(p)$ that heavily asymmetric-penalizes under-predictions. (5 Marks)

b) Explain the specific evolutionary mechanism within a Genetic Algorithm (GA) that allows it to escape the localized rugged traps that defeated the gradient-based LMA in this scenario. (5 Marks)

c) Contrast the GA approach with Particle Swarm Optimization (PSO). Conceptually, how does a GA's "Chromosome" explore the parameter space differently than a PSO "Particle" updating its velocity? (5 Marks)

d) **Python Implementation:** Write a Python function `def custom_draft_cost(simulated, observed):` that takes two equal-length arrays of force data and computes the total error using the specific asymmetric penalty logic you derived in part (a). (5 Marks)

Section C: System Dynamics and Structural Analysis (35 Marks)

Question 5: Causal Loop Paradoxes and Stability (15 Marks)
Consider the paradox of herbicide application: Applying chemical herbicide rapidly kills weeds, which increases crop yield in the short term. However, over multiple seasons, the surviving weeds develop genetic resistance, rendering the herbicide less effective and forcing the farmer to apply exponentially higher doses to maintain the same yield.

a) Formulate this scenario using Causal Loop Diagram (CLD) terminology. Explicitly define the intended Balancing loop and the unintended, delayed Reinforcing loop driving the paradox. (6 Marks)

b) Assume you formulate a state-space model for this ecosystem and evaluate the Jacobian matrix precisely at an equilibrium point of "Zero Weeds". If the dominant eigenvalue is calculated as $\lambda = +0.45$, what does the Stability Theorem mathematically prove regarding the long-term feasibility of this eradication strategy? (5 Marks)

c) If, alternatively, the eigenvalues were complex conjugates ($\lambda = -0.1 \pm 0.8i$), describe the physical trajectory and dynamic behavior of the weed population over multiple seasons. (4 Marks)

Question 6: Networked Hydrology and Stomatal Control (20 Marks)
You are designing a precision irrigation model for a genetically engineered drought-resistant crop. This specific crop exhibits a hyper-sensitive stomatal resistance (r_s) that instantly doubles upon a very minor drop in leaf water potential (Ψ_{leaf}).

a) Utilizing the electrical resistance analogy (Ohm's Law) for the Soil-Plant-Atmosphere Continuum (SPAC), mathematically derive what happens to the intermediate sap flow (q) and the internal stem potential (Ψ_{stem}) at the exact moment r_s doubles, assuming the deep soil potential (Ψ_{soil}) remains constant. (7 Marks)

b) How does this specific genetic modification fundamentally alter the Balancing feedback loop of the plant's water conservation strategy compared to a non-engineered crop? (6 Marks)

c) In the context of the FAO-56 Penman-Monteith equation, if this engineered crop is planted in a highly arid, extremely windy environment (characterized by a very low aerodynamic resistance r_a), mathematically justify whether the genetic modification (r_s) will exert a dominant or negligible impact on the total daily Evapotranspiration (ET). (7 Marks)

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